

# FormulANN Grand Prix

# Problem

$k$ -nearest neighbor queries:

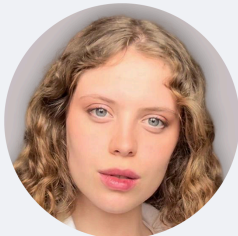
given a dataset  $S$  and a point  $q$ , find the  $k$  points in  $S$  closest to  $q$

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- 7 datasets
  - dimensions 384 - 3072
  - size 200k - 1.4M
- 1000 queries each
- query set
  - public for development
  - secret for the competition

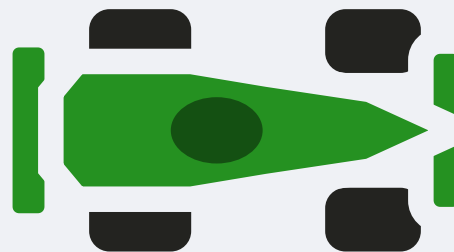
# Challenge prizes

- **Sherlock Holmes**: fastest getting average recall  $\geq 0.95$
- **Bianconiglio**: fastest getting average recall  $\geq 0.8$
- **Dory**: approach using the least amount of memory with recall  $\geq 0.95$
- **Marie Kondo**: fastest at building the index
- **Paperone**: the one using the fewest distance computations



# Friendly Neighborhood Solvers

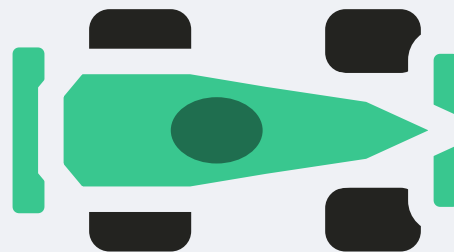
- Hierarchical k-means + quantization
- CPU+GPU (CUDA)
- C++



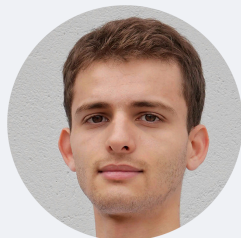
# Close Enough



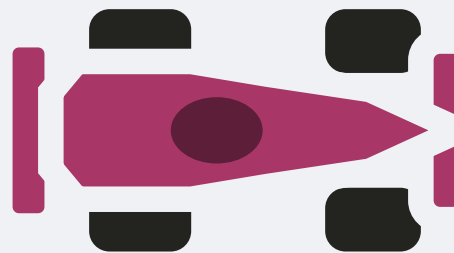
- HNSW + RaBitQ
- Tunes ef at build time, based on calibration queries
- Exact reranking
- CPU with AVX-512 instructions
- C++



# Kinda-neighbors



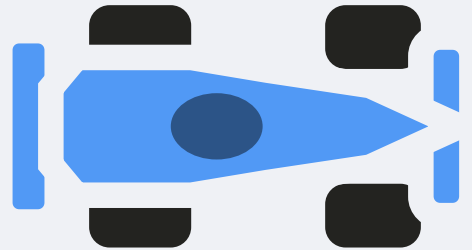
- HNSW with 8-bit scalar quantization
- Never computes a full distance
- Adds a patience parameter at the base HNSW layer
- Rust
- CPU



# ANNarchy



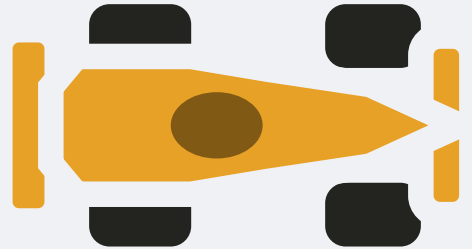
- HNSW or IVF, depending on the dataset
- No quantization
- CPU
- C++



# Neighbors grass



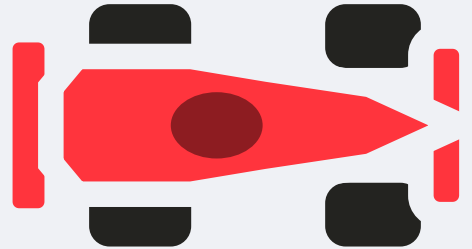
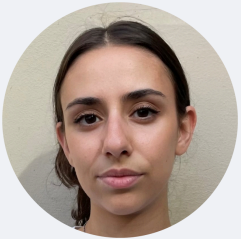
- HNSW + RaBitQ
- Never compute a full distance
- CPU with AVX512
- C++

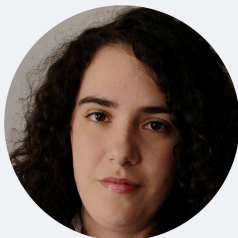


# ANNVedi



- Brute force over quantized representation (or HNSW as a fallback)
- Verify only a limited number of candidates
- GPU (CUDA)
- C++





# Finding NNemo

- IVF + SAQ quantization (SIGMOD26)
- No exact reranking
- CPU
- C++

